

Pillar Optimisation

Underground geotechnical — extraction ratio vs ground stability, AI-monitored

An interactive demo for the OreSight AI platform: an underground bord-and-pillar panel where the AI balances **ore recovery against ground stability** — sizing and sequencing pillars to maximise the extraction ratio while holding every pillar's factor of safety above target, fused with live geotechnical monitoring (convergence, stress, microseismic). The optimisation counterpart to the Safety demo, and OreSight's first step from surface into underground ground-control.

Visual model	Pillar FOS heat-map (plan-view pillar grid) as the hero, per the FSP/haul pattern
Domain	Bord-and-pillar panel, ~180 m depth, instrumented for convergence + microseismic
Context note	Pillars are an underground problem — framed as OreSight extending into underground coal (E. Kalimantan); adjust if a different setting is meant
Design system	Reuse OreSight light/dark control-room language from FSP
Sources	Lunder-Pakalnis / Salamon-Munro / Hedley-Grant pillar strength; Itasca, RocScience, Beck; Sintela / microseismic monitoring; ML pillar-stability literature
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1 · Context & goal

In bord-and-pillar (room-and-pillar) mining, ore is extracted by driving roadways and leaving rock **pillars** to hold up the overburden. The central trade-off is recovery vs stability: shrink the pillars and you win ore but lose **factor of safety (FOS)**; over-size them and you sterilise reserves. Pillar optimisation finds the highest safe extraction ratio — and modern practice fuses that design with **live monitoring** so the as-mined ground confirms the as-designed assumption.

Goal: a demo that *feels like one product* with FSP and the hauling demo — a live geotechnical screen where the AI does the engineering judgement (which pillars are at risk, how much more can be safely recovered, what support/sequence makes it safe) and re-assesses when conditions change.

What benchmark practice does (research synthesis)

- **Pillar strength is a width-to-height power law.** Standard formulas: **Lunder & Pakalnis** (hard rock), **Salamon & Munro** (coal), **Hedley & Grant** — $\text{strength} = K \cdot (w/h)^a$. Pillar **stress** from the **tributary-area** method. **FOS = strength / stress**; accepted targets **~1.6 for production pillars, 2.0 for barriers**.
- **The optimisation prize is real:** geometric optimisation has lifted recovery from **~44% to >80%** safely; W/H ratio is the master lever ($W/H < 0.6$ pillars are the problem children). Higher recovery levers: **backfill confinement**, and pillar-recovery methods (split-and-fender, pocket-and-wing, Christmas tree).
- **Monitoring stack (Sintela, Itasca, microseismic vendors):** roof-to-floor **convergence** meters, **extensometers** (in-rock displacement), **stress cells** and support **load cells**, and **microseismic** geophone arrays that triangulate fracturing *before* visible failure. Distributed fibre-optic sensing gives continuous coverage between geophones.
- **AI is proven here:** Random-Forest / SVM / ANN models predict pillar stability at **80–99% accuracy** from W/H, RMR and UCS; **hybrid frameworks** (empirical + numerical FLAC/RS2 + ML + real-time geophysics + probabilistic) improve predictions **15–20%** and target rockburst/subsidence risk.

2 · Metrics to adopt

North-star KPIs (large, top band)

Metric	Definition	Unit	Target band
Panel min FOS	Lowest pillar factor of safety (strength ÷ tributary stress) in the panel	ratio	≥1.6 prod; ≥2.0 barrier
Extraction ratio	Areal ore extracted vs in-situ; the recovery being optimised	%	maximise s.t. FOS (44→80%+)
Probability of failure	Panel instability likelihood from FOS distribution + monitoring trend	%	low single digits
Convergence rate	Worst-station roof-to-floor closure speed; the live ground-truth	mm/day	stable/decelerating

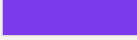
Supporting metrics (secondary band)

Metric	Definition / why it matters	Unit
W/H ratio	Pillar width ÷ mining height — the master strength lever; <0.6 is slender-risk	ratio
Stress / strength	Tributary stress as % of pillar strength (= 1/FOS); utilisation	%
Microseismic activity	Event rate, released energy, b-value — leading indicator of yielding	ev/h, J
RMR / UCS	Rock mass rating / uniaxial strength — key strength inputs	– / MPa
Roof span	Bord width vs stable-span limit; roof-fall governor in wide cuts	m
Support load	Bolt / prop load-cell reading vs capacity; is support taking design load	kN

Value model (reuse FSP approach): tonnes of additional *safe* recovery × margin, net of support/backfill cost and risk-adjusted exposure — derived, never canned.

3 · Pillar status model & colour coding

The FOS heat-map and the pillar table share one colour language; the bands key off factor of safety against the 1.6 production target.

Status	Meaning	Swatch
OK	$FOS \geq 1.6$ — within design	
WATCH	$1.3 \leq FOS < 1.6$ — reduced margin / monitor	
CRITICAL	$FOS < 1.3$ — support / no further extraction	
BARRIER	Barrier pillar (target $FOS \geq 2.0$) — outlined	
EXTRACTED	Mined void / roadway (bord)	
YIELDING	Microseismic / convergence flag overrides static FOS	

Key design idea: **monitoring overrides geometry**. When convergence is accelerating or microseismic energy is rising (b-value falling) on a pillar, it reads YIELDING even if its static FOS looks acceptable — that fusion of design and live data is the demo's intelligence.

4 · Screen layout & components

Follow the FSP page skeleton: breadcrumb → demo-head → KPI row → dark control-room screen → AI panel + side panels. Components top to bottom:

#	Component	Spec
1	KPI row	4 north-star cards: panel min FOS, extraction ratio, probability of failure, worst convergence rate. Up/down deltas vs plan.
2	Constraint (drum) strip	Reuse FSP bottleneck strip: binding constraint = pillar-strength / roof-span / convergence / seismic, with a 'governs now' shift indicator on re-assess.
3	Pillar FOS heat-map (HERO)	Plan-view grid of pillars coloured by FOS band; barrier pillars outlined; rooms shown as void; sensor glyphs on instrumented pillars. Click a pillar → detail card. SVG, same render approach as the FSP Gantt.
4	FOS × extraction trade-off curve	The optimisation story: a Pareto curve of FOS vs extraction ratio with the current point and the AI-recommended point; target-FOS floor line. Chart.js.
5	Scenario inject + free-form	Presets (mine deeper, weak seam band, pillar recovery, rainfall/pore pressure, nearby blast) + NL box → AI re-assesses FOS map and recovery. Mirror FSP.
6	Pillar table	Per-pillar rows: id, W/H, FOS, stress/strength %, convergence, status dot, AI action. Highlights at-risk and re-designed pillars.
7	Monitoring panels	Convergence trend per station; microseismic event-rate / energy strip (plan scatter or sparkline). FSP dark theme.
8	AI panel + risk gauge	renderAIResult-style: headline + recommendations + monitoringActions + value; a probability-of-failure gauge. Live on each re-assess.

Tabs (optional): **Stability map** · **Optimisation** (trade-off curve + table) · **Monitoring** — mirroring FSP's three tabs.

5 · AI capabilities

- **Stability assessment** — per-pillar FOS from W/H, RMR, UCS, depth; flag at-risk pillars (empirical + ML pattern).
- **Extraction-ratio optimisation** — the highest recovery that holds every pillar above target FOS; quantify the ore unlocked.
- **Design-vs-monitoring fusion** — weight live convergence / microseismic trends above static FOS when they disagree (the YIELDING override).
- **Support & recovery engineering** — recommend backfill confinement, split-and-fender / pocket-and-wing recovery, bolting or barrier re-design, each with the FOS it holds.
- **Scenario re-assessment** — deeper mining, weak seam, rainfall/pore-pressure, blast vibration → live re-solve with rationale (FSP-style).
- **Probabilistic risk** — probability of failure from the FOS distribution and monitoring trend, not FOS alone.
- **Monitoring tasking** — tell the geotech team which pillars to instrument next and why (links to the Safety demo).

6 · Claude prompt (drop-in for server.js)

House style (persona → exact JSON schema → domain rules → JSON-only). One endpoint `/api/pillar/analyze` handles 'assess + optimise current panel' and 'scenario re-assess' via an optional scenario. The `{headline, recommendations[], valueImpactUSD, narrative}` core reuses `renderAIResult`; specialised fields drive the bespoke UI. Ship a deterministic `pillarFallback()` so it runs with no API key.

```

PILLAR_PROMPT = `You are OreSight AI's geotechnical engineer for an underground
bord-and-pillar coal operation in East Kalimantan, Indonesia. A panel is developed at
~180 m depth in a 5.0 m seam; pillars sit on a regular grid with ~6.5 m roadways (bords).
Pillar strength follows a width-to-height power law (Salamon-Munro / Lunder-Pakalnis
style); pillar stress is estimated by the tributary-area method; FOS = strength / stress.
Target FOS >= 1.6 for production pillars, >= 2.0 for barrier pillars. The panel is
instrumented: roof-to-floor convergence stations, borehole extensometers, stress cells
and a microseismic geophone array.
&nbsp;
Given a JSON snapshot of a panel (pillar grid geometry & W/H ratios, rock-mass RMR/UCS,
depth, current extraction ratio, and live monitoring: convergence rates, stress-cell
loads, microseismic event rate/energy) - and optionally a scenario (deeper mining, weak
seam band, pillar recovery, rainfall/pore pressure, nearby blast) - assess stability and
optimise extraction. Return JSON with EXACTLY this structure:
&nbsp;
{
  "headline": "<conclusion: current panel FOS vs target and recovery opportunity>",
  "panelMinFOS": <number, lowest pillar factor of safety>,
  "extractionRatioPct": <number, current areal extraction>,
  "optimisedExtractionPct": <number, achievable while holding target FOS>,
  "bindingConstraint": "pillar-strength" | "roof-span" | "convergence" | "seismic",
  "constraintNote": "<why this governs, one line>",
  "atRiskPillars": [{ "id": "P-12", "fos": <n>, "whRatio": <n>,
    "stressStrengthPct": <n>, "status": "CRITICAL" | "WATCH" | "OK", "note": "<short>" }],
  "probabilityOfFailurePct": <0-100>,
  "recommendations": [{ "action": "<resize|sequence|backfill|support|destress>",
    "impact": "<quantified: FOS, recovery %, tonnes, US$>", "timeframe": "<when>" }],
  "monitoringActions": [{ "pillar": "P-12", "instrument": "convergence|extensometer|
    stress-cell|microseismic|inspection", "reason": "<short>" }],
  "valueImpactUSD": <integer, value of additional safe recovery net of cost, annualised>,
  "narrative": "<3-4 sentences linking geometry, monitoring and the FOS/recovery trade-off>"
}
&nbsp;
Domain rules:
- FOS = pillar strength / tributary-area stress; strength rises with W/H. Never recommend
  extraction that pushes any production pillar below FOS 1.6 or any barrier below 2.0.
- W/H ratio is the master geometric lever; flag W/H < 0.6 pillars as slender risks.
- Convergence acceleration and rising microseismic energy (falling b-value) are leading
  indicators of yielding - weight them ABOVE static FOS when they conflict.
- Higher-recovery levers: trim pillars only where FOS headroom allows; otherwise backfill
  confinement (raises strength), split-and-fender / pocket-and-wing recovery, or barrier
  re-design. State the FOS each option holds.
- Tie probabilityOfFailurePct to the FOS distribution AND the monitoring trend, not FOS
  alone.
- 3-5 atRiskPillars, 3-4 recommendations, 2-4 monitoringActions; each concrete & quantified.
&nbsp;
IMPORTANT: Return ONLY valid JSON, no other text.`;

```

Companion (optional, mirrors FSP): a small PILLAR_PARSE_PROMPT behind /api/pillar/parse turns a free-text note ('weak band in P-12 to P-15, water making') into a structured scenario the engine re-assesses.

7 · Design system (reuse OreSight tokens)

FOS OK / brand	Watch	Critical	Barrier / sensor	Yielding	Control-room
#15803d	#b45309	#b91c1c	#0e7490	#7c3aed	#10150f

- **Type:** Inter (UI) + JetBrains Mono (labels/data), already loaded.
- **Reuse verbatim:** kpi-card, fsp-screen / fsp-tab / fsp-status, bottleneck-strip, ai-panel + renderAIResult, risk-item, copilot, freeform-row, status-dot.
- **New components:** pillar FOS heat-map (SVG plan grid), FOS legend, FOS×extraction trade-off curve, convergence trend, microseismic strip, probability-of-failure gauge.
- **Motion:** on re-assess, re-designed pillars recolour with a transition and the constraint strip updates the 'governs now' shift — same feel as FSP block re-shuffle.

8 · Build notes

- **Page:** mining-demo/public/pillar.html + js/pillar.js; **endpoint:** /api/pillar/analyze (+ optional /api/pillar/parse) in server.js; **fallback:** data/pillar.js (deterministic).
- **Deterministic engine is source of truth** for the on-screen FOS map (always re-solves visibly); AI parses free-text and writes rationale on top — works fully with no API key.
- **Matrix wiring:** add a 'Pillar / Ground-Control Optimisation' cell (geotech / mine-development horizon) on the value-chain landing page, LIVE DEMO, linking to /pillar.html.
- **Nav:** add a 'Pillar' link in the shared nav; cross-link from the Safety demo (shared geotech narrative).

Sources

- Pillar design & strength formulas (Lunder-Pakalnis, Salamon-Munro, Hedley-Grant; FOS 1.6/2.0) — sciELO.org.za (hard-rock bord-and-pillar); ijirt.org IJIRT153013; link.springer.com 10.1007/s44288-025-00295-3
- Toward pillar design to prevent collapse of room-and-pillar mines — stacks.cdc.gov/view/cdc/9341
- Extraction-ratio / recovery optimisation (44→80%; W/H, split-and-fender, pocket-and-wing) — sciencedirect.com S2095268618301265; link.springer.com 10.1007/s12665-023-10801-w; sciELO.org.za (backfill confinement)
- Monitoring: convergence, extensometers, stress/load cells, microseismic, fibre-optic — sintela.com; sciencedirect.com S2467967417300077 (microseismic, retreat room-and-pillar)
- ML / hybrid pillar-stability prediction (RF/SVM/ANN 80–99%; hybrid +15–20%) — sciencedirect.com S1474706525003857; ncbi.nlm.nih.gov PMC8871988 (logistic model trees); link.springer.com 10.1007/s00603-025-04841-w (cave-pillar ML)
- Numerical modelling context — Itasca (FLAC3D/3DEC), RocScience (RS2/Examine), Beck Engineering, Map3D